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Hausdorff Measure of Critical Values

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*Alla mia famiglia,
vi voglio immensamente bene.*

*A tutte le persone speciali che mi hanno sostenuta sempre,
e ai piccoli grandi momenti insieme.*

*A Wlad, avrei voluto condividere tutto questo con te.
Mi manchi ogni giorno.*

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Introduction

In 1939 Anthony Morse published a paper [17] proving that a C^k map f from a differentiable manifold M of dimension m into a differentiable manifold N of dimension $n = 1$, with $k \geq m + n - 1$, sends the set of singularity points (points at which the Jacobian of f has not maximum rank) into a set of Lebesgue measure zero in N .

A counterexample for the case $k \leq m + n - 1$ was found in a paper published by Hassler Whitney in 1935 [25], using a C^1 function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with null derivative at many points but not constant.

Three years later, in 1942, Arthur Sard published another paper [22], showing that the same result applied with a generic integer n .

In 1965 Sard then published a refinement of this theorem [24]. This paper helped to understand even better how critical points for a map from a finite-dimensional Banach manifold into a Banach manifold behave. Naming A_r the set of points for which the Jacobian of the map f has rank $a \leq r$, Sard proved that, if f is smooth enough, $f(A_r)$ has dimension at most r .

Ivan Kupka proved later in 1965, in the paper [15], that the hypothesis of M being finite-dimensional cannot be removed, building a counterexample on the infinite-dimensional case.

The Morse-Sard theorem became an important result to keep under consideration when working with manifolds or Banach manifolds: it states that when we work with a smooth enough map from a manifold/Banach manifold, the image is almost everywhere a manifold, which is very useful.

In this thesis we are going to go over the paper [24]. Our plan is to first define all of the objects useful to the proof, including Banach manifolds — we will also use and define Hausdorff measure and dimension and some notions of functional and real analysis —, list and prove all of the key lemmas and then get into the main proof, separating it into four parts. This will allow us to follow the different parts without getting lost, adding some details in the process.

In the end of the thesis we will talk about some more results and refinements that followed the paper [24].

Chapter 1

Preliminaries

As mentioned in the Introduction, in this chapter we will provide the necessary definitions to proceed with the lemmas and main proof discussed in this thesis.

We will first quickly recall the definition of differentiable manifolds and then define Banach manifolds, continue with some auxiliary functions useful to the main proof, and conclude this chapter by recalling Hausdorff measure and dimension definition and introducing some notation.

1.1 Differentiable manifolds

We recall the definition of differentiable manifolds (see [16, §1]; see also [19]).

Definition 1. A **n -dimensional differentiable manifold of class C^k** is a couple $(X, \mathcal{A})$ where X is a Hausdorff separable topological space with topology τ_X and $\mathcal{A}$, atlas of class C^k on X , is a collection of charts (U_i, ϕ_i) , $\phi : U_i \rightarrow \mathbb{R}^n$ homeomorphism such that

- $\bigcup_{i \in I} U_i = X$;
- for each $i, j \in I$, $\phi_i \circ \phi_j^{-1} : \phi_j(U_i \cap U_j) \rightarrow \phi_i(U_i \cap U_j)$ is of class C^k .

The integer n is called dimension of X .

1.2 Banach manifolds definition

We now proceed to define Banach manifolds (see [20]; see also [16, §3]):

Definition 2. Let E and F be two Banach spaces, $U \subset E$ open set, $f : U \rightarrow F$. We say that f is **differentiable** if there exists a map $Df : U \rightarrow \mathcal{L}(E, F)$ such that $\forall x \in U$

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Df(x)(h)\|_F}{\|h\|_E} = 0.$$

The map Df is called **differential** of f . If Df is continuous, f is of class C^1 . We set inductively

$$D^k f := D(D^{k-1} f) : U \rightarrow \mathcal{L}^k(E, F) \cong \mathcal{L}(E, \mathcal{L}^{k-1}(E, F)).$$

If $D^k f$ is continuous, f is of class C^k . If f is C^k for all $k \in \mathbb{N}$, we say that f is C^∞ .

Definition 3. Let E be a Banach space, $k \in E^m$ a linear function, $k \neq 0$. Then we call $\{v \in E : k(v) \geq 0\}$ **half Banach space**, and $\partial H := \{v \in E : k(v) = 0\}$ the **boundary** of H .

Definition 4. Let $H \subset E$ be a half Banach space, $O \subset H$ an open set of H in the subspace topology. Let F be a Banach space, and $f : O \rightarrow F$ a map, $p \in \partial H \cap O$. We say that f is **C^k at p** if there exists an open set $U \subset E$ such that $p \in U$ and a C^k map $g : U \rightarrow F$ such that $g|_{O \cap U} = f|_{O \cap U}$.

Definition 5 (Banach charts). Let (X, τ_X) be a Hausdorff topological space. A **chart** on X is a triplet (U, ϕ, E) , where $U \in \tau_X$ is an open subset of X , E is a Banach space or a half Banach space and $\phi : U \rightarrow \phi(U) \subset E$ is a homeomorphism onto an open subset of E .

Definition 6 (Banach Atlases). Let (X, τ_X) be a Hausdorff topological space. A **Banach Atlas of class C^k on X** , with $\mathbb{N} \ni q \geq 1$ is a collection $\{(U_i, \phi_i, E_i)\}_{i \in I}$ of Banach charts on X such that

- $\bigcup_{i \in I} U_i = X$;
- for each $i, j \in I$, $\phi_i \circ \phi_j^{-1} : \phi_j(U_i \cap U_j) \rightarrow \phi_i(U_i \cap U_j)$ is of class C^k .

Definition 7 (Banach Manifold). A **Banach manifold of class C^k** is a couple $(X, \mathcal{A})$ where X with its topology τ_X is a Hausdorff separable topological space and $\mathcal{A}$ is a Banach Atlas of class C^k on X .

1.3 Useful auxiliary functions

We now define the auxiliary functions mentioned beforehand and characterize them.

Definition 8. Define the function $c : \mathbb{N} \times \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ by

$$c(k, \rho) = \begin{cases} 1 & 0 \leq k < \rho; \\ 2 + \frac{k(k - \rho)}{\rho} & \rho \leq k. \end{cases} \quad (1)$$

Definition 9. Define the second auxiliary function as $\nu : \mathbb{N} \times \mathbb{R} \rightarrow \mathbb{R}_{>0}$, where $\nu(k, \rho)$ is the smallest integer strictly greater than k/ρ :

$$\nu(k, \rho) - 1 \leq k/\rho < \nu(k, \rho). \quad (2)$$

Proposition 1. For all $k \in \mathbb{N}$, $\rho \in \mathbb{R}_{>0}$ the following inequalities hold:

$$c(k+1, \rho) \geq c(k, \rho) \geq 1; \quad (3)$$

$$c(k, \rho) \geq \nu(k, \rho); \quad (4)$$

$$c(k, \rho) \geq c(k-1, \rho) + \nu(k, \rho) - 2 \quad \text{whenever } 0 < \rho \leq k \text{ and } \nu(k, \rho) \geq 3; \quad (5)$$

$$c(k, \rho) \geq c(k+1, \rho+1) \quad \text{whenever } 0 < \rho \leq k. \quad (6)$$

Proof. (3) follows from monotonicity and positivity of k and ρ : if $0 \leq k < \rho$ and $0 \leq k+1 < \rho$ then $c(k+1, \rho) = 1 = c(k, \rho)$; if $0 \leq k < \rho$ but $0 < \rho \leq k+1$, then

$$c(k+1, \rho) = 2 + \frac{(k+1)(k+1-\rho)}{\rho} \geq 2 > c(k, \rho) = 1.$$

Lastly if $0 \leq \rho < k$, then $0 \leq \rho < k$ and the result follows by monotonicity of c in k .

(4) is similarly checked: if $0 \leq k < \rho$ then $c(k, \rho) = 1 > k/\rho$; if $\rho \leq k$, we prove that $c(k, \rho) \geq k/\rho + 1$. The result follows since $k/\rho + 1 \geq \nu(k, \rho)$ by definition of ν :

$$2 + \frac{k(k-\rho)}{\rho} \geq 1 + \frac{k}{\rho} \Leftrightarrow \frac{k(k-\rho)}{\rho} \geq \frac{k-\rho}{\rho},$$

which is true, since $k \in \mathbb{N}$, $k \geq \rho > 0 \Rightarrow k \geq 1$.

(5) can be obtained by algebraic manipulations:

$$\begin{aligned}
c(k, \rho) &= 2 + \frac{k(k - \rho)}{\rho} \\
&\geq 2 + \frac{k(k - \rho)}{\rho} - \frac{k - 1}{\rho} \\
&= 2 + \frac{k(k - \rho)}{\rho} - \frac{k - 1}{\rho} - \frac{k}{\rho} + \frac{k}{\rho} + 2 - 2 \\
&= 2 + \frac{(k - 1)(k - 1 - \rho)}{\rho} + \frac{k}{\rho} + 1 - 2 \\
&\geq c(k - 1, \rho) + \nu(k, \rho) - 2.
\end{aligned}$$

To conclude (6) follows by some more algebraic manipulations:

$$\begin{aligned}
c(k, \rho) &\geq c(k + 1, \rho + 1) \\
&\Leftrightarrow \frac{k(k - \rho)}{\rho} \geq \frac{(k + 1)(k - \rho)}{\rho + 1} \\
&\Leftrightarrow (\rho + 1)k(k - \rho) \geq \rho(k + 1)(k - \rho) \\
&\Leftrightarrow \rho k^2 - \rho^2 k + k^2 - \rho k \geq \rho k^2 - \rho^2 k + \rho k - \rho^2 \\
&\Leftrightarrow k^2 - 2\rho k + \rho^2 \geq 0 \\
&\Leftrightarrow (k - \rho)^2 \geq 0,
\end{aligned}$$

which is always true, as desired. $\square$

1.4 Hausdorff measure, some notation

We first recall the definition of Hausdorff measure and dimension (see [9]; see also [16, § 1]), and then set some notation.

Definition 10 (Hausdorff measure). Let E be a Banach space and B a subset. For every $\delta > 0$, $s > 0$, define

$$\mathcal{H}_\delta^s(B) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(U_i)^s : \mathcal{P}(E) \supset \{U_i\}_{i \in \mathbb{N}}, \bigcup_{i \in \mathbb{N}} U_i \supset B, \text{diam}(U_i) < \delta \ \forall i \in \mathbb{N} \right\}.$$

If the set over which we take the inf is empty we define $\mathcal{H}_\delta^s(B)$ to be $+\infty$.

As δ decreases $\mathcal{H}_\delta^s(B)$ increases, therefore we can take the limit as $\delta \rightarrow 0$, and define the **s-dimensional Hausdorff measure** accordingly:

$$\mathcal{H}^s(B) = \sup_{\delta > 0} \mathcal{H}_\delta^s(B) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s(B).$$

Proposition 2. Let E be a Banach space, $F \subset E$. Then

$$\inf \{s \geq 0 : \mathcal{H}^s(F) = 0\} = \sup \{s \geq 0 : \mathcal{H}^s(F) = \infty\}.$$

Proof. Consider $\{U_i\}_{i \in \mathbb{N}}$, $\cup_{i \in \mathbb{N}} U_i \supset B$, $\text{diam}(U_i) < \delta \ \forall i \in \mathbb{N}$ and $0 \leq t < s$. Then

$$\sum_{i=1}^{\infty} \text{diam}(U_i)^s \leq \delta^{s-t} \sum_{i=1}^{\infty} \text{diam}(U_i)^t.$$

By taking infima we obtain

$$\mathcal{H}_\delta^s(F) \leq \delta^{s-t} \mathcal{H}_\delta^t(F).$$

The thesis follows by letting δ decrease to 0, since

$$\begin{aligned} \mathcal{H}^t(F) < \infty &\Rightarrow \mathcal{H}^s(F) = 0 \\ \mathcal{H}^t(F) = \infty &\Leftarrow \mathcal{H}^s(F) > 0. \end{aligned}$$

This means that $\mathcal{H}^s(F)$ skips from ∞ to 0 assuming at most one real positive value (figure 4.2). $\square$

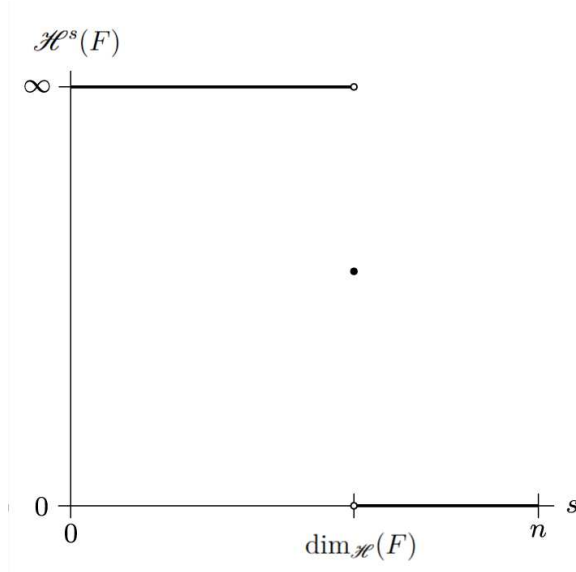


Figure 1: Hausdorff measure $\mathcal{H}^s(F)$ as a function of $s \geq 0$ for any fixed set $F \subset E$ of a Banach space

Definition 11 (Hausdorff dimension). Let E be a Banach space, $F \subset E$. The Hausdorff dimension of the set F is defined as follows:

$$\dim_{\mathcal{H}}(F) := \inf \{s \geq 0 : \mathcal{H}^s(F) = 0\} = \sup \{s \geq 0 : \mathcal{H}^s(F) = \infty\}.$$

Remark. Even though this is not obvious at first glance, Hausdorff dimension has properties that are coherent with an intuitive idea of dimension; in particular, one can show that

- if E is a Banach space and $A \subset E$ is such that A is at most countable, then $\dim_{\mathcal{H}}(A) = 0$;
- it has properties of monotonicity: if $A \subset B \subset E$, E Banach space, then $\dim_{\mathcal{H}}(A) \leq \dim_{\mathcal{H}}(B)$;
- It behaves well under countable unions: if E is a Banach space and $\{A_i\}_{i \in \mathbb{N}} \subset \mathcal{P}(E)$, then $\dim_{\mathcal{H}}(\cup_{i \in \mathbb{N}} A_i) = \sup_{i \in \mathbb{N}} \dim_{\mathcal{H}}(A_i)$;
- If E is an n -dimensional Banach space, $0 \leq n \leq \infty$, and $A \subset E$ is an open subset, then $\dim_{\mathcal{H}}(A) = \dim_{\mathcal{H}}(E) = n$;
- If N is an n -dimensional Banach manifold, $0 \leq n \leq \infty$, then $\dim_{\mathcal{H}}(N) = n$.

These properties are not the object of this thesis: we will neither prove nor use them; to read more about them, see [9].

Notation. We say that $B \subset E$ Banach space is **s-positive**, **s-finite**, **s-null** or **s-infinite** according to $\mathcal{H}^s(B)$ having the respective property. We say that B is *s-sigmafinite* if B can be written as the countable union of s -finite sets.

Let $f : M \rightarrow N$ be a map from a m -dimensional differentiable manifold M into a Banach manifold N , f, M, N are of class C^k , $1 \leq k \leq \infty$. **The rank** r of a point $x \in M$ is defined as the rank of the tangent map $Df(x)$; that is, r is the dimension of the image $Df(x)E$. $k = m - r$ may be called corank of x .

Suppose that $s > 0$ is a real number and N a Banach manifold, $S \subset N$. We say that S is **s-null** if there exist countably many charts $(V_i, \psi_i, Y_i)_{i \in \mathbb{N}}$ of N such that $B \subset \cup_{i \in \mathbb{N}} V_i$ and $\psi_i(B \cap V_i)$ is s -sigmafinite for all $i = 1, 2, \dots$. We say that B is **s-positive** if B is not s -null.

Chapter 2

Technical tools

In this chapter, we will work through six lemmas. Each one will help us understand how the properties observed in the notation above behave under unions, intersections, diffeomorphisms, and build up toward Sard's theorem.

Throughout this chapter, let E be a one dimensional Banach space (isometric to $\mathbb{R}$); F, G Banach spaces, and M, N respectively C^p, C^q Banach manifold, with $1 \leq p, q \leq \infty$.

Lemma 1. Suppose that $\theta : \Omega \subset F \rightarrow G$ is a differentiable map of an open subset Ω of F into G . If $A \subset \Omega$ is s -null or s -sigmafinite, then so is $\theta(A)$.

Proof. For each pair ν, λ of positive integers, let $\Omega^{\nu, \lambda}$ be the set of points $x \in \Omega$ such that

$$\|\theta(x') - \theta(x)\| < \nu \|x' - x\| \quad \text{whenever} \quad \|x' - x\| < \frac{1}{\lambda}, \quad x' \in \Omega. \quad (7)$$

There are countably many such sets, and

$$\Omega = \bigcup_{\nu, \lambda} \Omega^{\nu, \lambda},$$

Now, if A is s -null, so is its subset $A \cap \Omega^{\nu, \lambda}$, and therefore $\theta(A \cap \Omega^{\nu, \lambda})$: by (7), for each $B \subset A \cap \Omega^{\nu, \lambda}$ such that $\text{diam}(B) < \frac{1}{\lambda}$,

$$\text{diam}(\theta(B)) \leq \nu \text{diam}(B).$$

This implies that for each $\{B_i\}_{i \in \mathbb{N}} \subset \mathcal{P}(A \cap \Omega^{\nu, \lambda})$

$$\sum_{i \in \mathbb{N}} \text{diam}(\theta(B_i))^s \leq \nu^s \sum_{i \in \mathbb{N}} \text{diam}(B_i)^s;$$

which means that for all $0 < \delta < \lambda$

$$\mathcal{H}_\delta^s(\theta(A \cap \Omega^{\nu, \lambda})) \leq \nu^s \mathcal{H}_\delta^s(A \cap \Omega^{\nu, \lambda}) = 0.$$

and therefore taking the limit for $\delta \rightarrow 0$

$$\mathcal{H}^s(\theta(A \cap \Omega^{\nu, \lambda})) \leq \nu^s \mathcal{H}^s(A \cap \Omega^{\nu, \lambda}) = 0.$$

To conclude, we note that if for all ν, λ , $\theta(A \cap \Omega^{\nu, \lambda})$ is s -null, so is $\theta(A) = \cup_{\nu, \lambda} \theta(\Omega^{\nu, \lambda} \cap A)$, by subadditivity of the s -dimensional Hausdorff measure.

The proof for A s -sigmafinite is similar. $\square$

Observation 1. Working with local charts, Lemma 1 easily implies the following result.

Lemma 1'. Suppose that $\theta : \Omega \subset N \rightarrow M$ is a C^q map, $q \geq 1$, of an open subset Ω of a Banach manifold into a Banach manifold. If $A \subset \Omega$ is s -null or s -sigmafinite, then so is $\theta(A)$.

Lemma 2. Any subset of an s -null set on N is s -null. Likewise, a subset of an s -sigmafinite set is s -sigmafinite.

Proof. It is immediate: if $B \subset A \subset N$ is such that A is s -null, there exist a countable set of Banach charts on N

$$\{(V_i, \phi_i, Y_i)\}_{i \in \mathbb{N}}, \mid \forall i \in \mathbb{N}, \phi_i(A \cap V_i) \text{ is } s\text{-null}$$

The same set of Banach charts proves that B is s -null, as $B \subset A$ and $B \cap V_i \subset A \cap V_i$. $\square$

Lemma 3. A countable union of s -null sets in N is s -null. Likewise, a countable union of s -sigmafinite sets is s -sigmafinite.

Proof. Suppose that

$$B = \bigcup_{\nu \in \mathbb{N}} B^\nu \subset N$$

and that charts $\{(V^{\nu, i}, \phi^{\nu, i}, Y^{\nu, i})\}_{\nu, i \in \mathbb{N}}$ exist such that

$$B^\nu \subset \bigcup_{i \in \mathbb{N}} V^{\nu, i}, \quad \psi^{\nu, i}(B^\nu \cap V^{\nu, i}) \text{ is } s\text{-null} \quad \forall \nu \in \mathbb{N}.$$

Then $B \subset \cup_{\nu, i} V^{\nu, i}$ and therefore all we have to prove is that $\forall \nu \in \mathbb{N}$, $\psi^{\nu, i}(B \cap V^{\nu, i})$ is s -null. In this way, the set of charts $\{(V^{\nu, i}, \phi^{\nu, i}, Y^{\nu, i})\}_{\nu, i \in \mathbb{N}}$ will be exactly the one sufficient to conclude.

To prove that $\psi^{\nu, i}(B \cap V^{\nu, i})$ is s -null, we will demonstrate that $\psi^{\nu, i}(B^\lambda \cap V^{\nu, i})$ is s -null, with $\lambda \in \mathbb{N}$. Then, by subadditivity of the Hausdorff measure on Banach

spaces, $\psi^{\nu,i}(B \cap V^{\nu,i}) = \cup_{\lambda \in \mathbb{N}} \psi^{\nu,i}(B^\lambda \cap V^{\nu,i})$ will also be s -null.

To do so, we shall show that $\psi^{\nu,i}(B \cap V^{\nu,i} \cap V^{\lambda,j})$ is s -null. This is sufficient by the same argument aforementioned, since

$$B^\lambda \cap V^{\nu,i} \subset \bigcup_{j \in \mathbb{N}} B^\lambda \cap V^{\nu,i} \cap V^{\lambda,j}.$$

We therefore consider $\psi^{\nu,i}(B^\lambda \cap V^{\nu,i} \cap V^{\lambda,j})$. Now, $\psi^{\lambda,j}(\psi^{\nu,i})^{-1}$ is a diffeomorphism of $\psi^{\nu,i}(V^{\nu,i} \cap V^{\lambda,j})$ onto $\psi^{\lambda,j}(V^{\nu,i} \cap V^{\lambda,j})$; and $\psi^{\lambda,j}(B^\lambda \cap V^{\lambda,j})$ is s -null by construction. Hence, $\psi^{\lambda,j}(B^\lambda \cap V^{\nu,i} \cap V^{\lambda,j})$ is s -null by Lemma 2 and $\psi^{\nu,i}(B^\lambda \cap V^{\nu,i} \cap V^{\lambda,j})$ is s -null by Lemma 1. $\square$

Lemma 4. If $B \subset N$ is s -sigmafinite, then B is $(s + \rho)$ -null for all $\rho > 0$.

Proof. This is also immediate, as the same property applies to Banach spaces: consider $A \subset E$. If A is s -sigmafinite, there exist $\{A_i\}_{i \in \mathbb{N}}$ such that $A \subset \cup_{i \in \mathbb{N}} A_i$ and $\mathcal{H}^s(A_i) < \infty \ \forall i \in \mathbb{N}$. This means that $\dim_{\mathcal{H}}(A_i) \leq s$, and hence $\mathcal{H}^{s+\rho}(A_i) = 0 \ \forall \rho > 0, \rho \in \mathbb{R}$, as desired.

Moving to Banach manifolds, being B s -sigmafinite, we derive that there exist a countable set of charts $\{(V_i, \psi_i, Y_i)\}_{i \in \mathbb{N}}$ such that $\forall i \in \mathbb{N}$, $\psi_i(B \cap V_i)$ is s -sigmafinite. This means that $\psi_i(B \cap V_i)$ is also $(s + \rho)$ -null for all $\rho > 0, \rho \in \mathbb{R}$, and that $B \subset N$ is $(s + \rho)$ -null. $\square$

Lemma 5. If $B \subset N$ and if one chart (V, ψ, Y) for N exists such that $\psi(B \cap V)$ is s -positive in Y , then B is s -positive in N .

Proof. Otherwise, B would be s -null in N , and $B \cap V$ also, by Lemma 2. Then $\psi(B \cap V)$ would be s -null in Y , by Lemma 1', which is a contradiction. $\square$

Remark. Besicovitch [1] has shown that if $B \subset N$ has dimension $n \leq s$, B is not necessarily s -sigmafinite.

Lemma 6. Consider a C^q map θ of an open set $\Theta \subset E^{r+k} = E^r \times E^k$ into a Banach space Y , where

$$1 \leq k < \infty, \quad 0 \leq r < \infty, \quad 1 \leq q \leq \infty.$$

Choose an arbitrary set of axes in E^k , and suppose that

$$v_0 \in E^r, \quad w_0 = (w_0^1, \dots, w_0^k) \in E^k, \quad (v_0, w_0) \in \Theta,$$

$$\theta(v_0, w_0) = 0, \quad D_2\theta(v_0, w_0) \neq 0,$$

where D_2 indicates the partial derivative relative to $w \in E^k$ when $v \in E^r$ is fixed. Then there is at least one coordinate in E^k , say the k -th, for which the following holds. There exists an open neighborhood Ω of $(v_0, w_0) \in E^{r+k}$, an open neighborhood $\tilde{\Omega}$ of $(v_0, w_0^1, \dots, w_0^{k-1}) \in E^{r+k-1}$, and a C^q map η of $\tilde{\Omega}$ into E^1 such that

$$\theta(v, w) = 0, \quad (v, w) \in \Omega, \tag{8}$$

implies that

$$w^k = \eta(v, w^1, \dots, w^{k-1}), \quad (v, w^1, \dots, w^{k-1}) \in \tilde{\Omega} \tag{9}$$

The lemma does not assert that (9) implies (8). This lemma, adequate for our purposes, may be strengthened to allow E^r to be an arbitrary Banach space.

Proof. Suppose first that $Y = E^1$. By hypothesis, at least one partial derivative of θ with respect to a component of $w \in E^k$ must be $\neq 0$. Say that $\frac{\partial \theta}{\partial w^k} \neq 0$.

Then by the Implicit Function Theorem, there exist open neighborhoods Ω , $\tilde{\Omega}$ and a C^q map η such that (8) implies (9), and in this case, (9) implies (8).

Now, suppose that Y is an arbitrary Banach space. By hypothesis, the map $D_2\theta(v_0, w_0) \neq 0$. Therefore an element $w_* \in E^k$ exists such that

$$D_2\theta(v_0, w_0)(w_* - w_0) \neq 0 \in Y.$$

Now, put

$$E^1 = \text{span } D_2\theta(v_0, w_0)(w_* - w_0),$$

and put

$$\theta_* = P\theta$$

where P is a linear continuous projection onto E^1 . Such projections surely exist since E^1 is one-dimensional (we have a topological complement since E^1 is finite-dimensional, which means we also have a projection, see [4]; see also [8]).

Then (8) implies that

$$\theta_*(v, w) = 0.$$

Also

$$D_2\theta_*(v_0, w_0) = PD_2\theta(v_0, w_0) \neq 0,$$

since

$$PD_2\theta(v_0, w_0)(w_* - w_0) = D_2\theta(v_0, w_0)(w_* - w_0).$$

Now θ_* is a map into E^1 . By the first two paragraphs of the proof. Then Ω , $\tilde{\Omega}$ and η exist such that

$$\theta_*(v, w) = 0, \quad (v, w) \in \Omega$$

implies (9). A fortiori (8) implies (9), and the proof is complete.

□

Chapter 3

Main Theorem

We now begin exploring the central piece of paper [24]. As mentioned in the Introduction, we will split the proof into four parts, adding some observations and the strategy beforehand.

3.0 The theorem

Theorem 1. Suppose that f is a C^q map of a C^q class manifold M of dimension $m < \infty$ with a countable basis into a C^q Banach manifold N , where $1 \leq q \leq \infty$.

Let A_r denote the set of points of M where Df has rank at most r , $r = 0, 1, 2, \dots$. Suppose $\rho > 0$ is a real number, and let $k = m - r$, $k \in \mathbb{N}$. Let $c : \mathbb{N} \times \mathbb{R}_{>0}$ be the function defined in (1). Then $f(A_r)$ is $r + \rho$ -null if

$$q \geq c(k, \rho).$$

In particular, if f is C^∞ then $f(A_r)$ has dimension at most r .

3.1 First observations and strategy

Observations. We can assume M is homogeneous: otherwise, we can simply apply the theorem to every homogeneous subset of M , since $q \geq c(k, \rho)$ implies $q \geq c(k - 1, \rho)$ by inequality (5).

We observe that, since M has a countable basis, M is m -sigmafinite, so $f(M)$ is also m -sigmafinite by Lemma 1'. By Lemmas 4 and 2 and it follows that $f(M)$ and $f(A_r)$ are $(r + \rho)$ -null if $r + \rho > m$. From now on we will restrict to the case $r + \rho \leq m$.

Strategy. We define B_r to be the set of points of M of rank r . In this way

$$A_r = B_0 \cup B_1 \cup \dots \cup B_r, \quad r = 0, 1, \dots, m.$$

To prove that $f(A_r)$ is $(r + \rho)$ -null whenever $q \geq c(k, \rho)$ it will suffice to prove that $f(B_r)$ is $(r + \rho)$ -null whenever $q \geq c(k, \rho)$. In that case we get $q \geq c(k, \rho) \geq c(k + 1, \rho + 1)$: the first inequality follows by the hypothesis and the second from (6).

Then substituting r with $r - 1$ and ρ with $\rho + 1$, we get that B_{r-1} is $(r - 1 + \rho + 1)$ -null, that is, $(r + \rho)$ -null. Similarly, by iterating, we get that $B_0, \dots, B_{r-2}$ are all $(r + \rho)$ -null. Since

$$f(A_r) = f(B_0) \cup f(B_1) \cup \dots \cup f(B_r),$$

it follows that $f(A_r)$ is $(r + \rho)$ -null.

To prove that $f(B_r)$ is $(r + \rho)$ -null it will be sufficient to show that each point of B_r lies in an open set Ω such that $f(B_r \cap \Omega)$ is $(r + \rho)$ -null: countably many such open sets will then cover B_r by the countability of M 's basis.

From this point on, the proof will be divided into four parts.

3.2 Local analysis of f

Our goal in this first part of the proof is to build some useful charts and maps that will help us later on in the proof, and see how they transform B_r sets.

Consider a point $u_0 \in B_r$ and suppose that $f(u_0)$ is not a boundary point of N . We choose charts (U, ϕ, X) for M and (V, ψ, Y) for N such that:

$$u_0 \in U, \quad \phi(u_0) = 0 \in X = E^m;$$

$$f(u_0) \in V, \quad \psi(f(u_0)) = 0 \in Y.$$

Here, E^m is the m -dimensional Banach space (unique up to isomorphism); the sets U, V are open in M, N respectively, and ϕ and ψ are C^q diffeomorphisms onto $\phi(U), \psi(V)$ open sets in X, Y respectively.

We define the function g as follows:

$$g = \psi \circ f \circ \phi^{-1}.$$

In this way g is a C^q class map of $\phi(U)$ onto $\psi(V)$, with $q \geq 1$ by (3). Now let

$$t = Dg(0)$$

be the map tangent to g at 0. We put

$$Y_1 = t(X) \subset Y.$$

Since t is a linear map and u_0 is of rank r , Y_1 is a linear space of dimension $r \leq m < \infty$. It follows that there exists a topological complement Y_2 of Y_1 in Y (see [4]; see also [8]). Then we can write each $y \in Y$ as

$$y = y_1 + y_2, \quad y_1 \in Y_1, \quad y_2 \in Y_2$$

in one unique way; in this way we can define $Q_1 : Y \rightarrow Y_1$, $Q_2 : Y \rightarrow Y_2$ by the relations:

$$Q_1(y) = y_1, \quad Q_2(y) = y_2.$$

These two operators are linear and continuous by the definition of topological complement.

Let X_2 be the kernel of t and define X_1 to be $X_2^\perp$, let $\tau = t|_{X_1}$ be the restriction of t to X_1 . In this way

$$\tau(X_1) = Y_1 = t(X), \quad X_1 = \tau^{-1}(Y_1) \quad t(X_2) = 0.$$

We deduce that X_1 , Y_1 are linearly homeomorphic and

$$\dim(X_1) = \dim(Y_1) = r, \quad \dim(X_2) = m - r = k.$$

We now introduce a new chart for M near u_0 . To this end, we define $\sigma : \phi(U) \subset X \rightarrow X$ as follows:

$$\sigma(w) = (x_1, x_2), \quad w \in \phi(U),$$

with

$$\begin{aligned} x_1 &= \tau^{-1}Q_1g(w) \in X_1; \\ x_2 &= \text{Proj}_{X_2}(w) \in X_2. \end{aligned} \tag{10}$$

Note that σ is well defined, since $g(w) \in Y$ domain of Q_1 and $Q_1 g(w) \in Y_1$ domain of τ^{-1} . The map σ is of class C^q since τ^{-1} , Q_1 , Proj_{X_2} are linear continuous operators.

The map $D_\sigma(0)$ tangent to σ at 0 carries $w' \in X$ into (x'_1, x'_2) , where

$$\begin{aligned} x'_1 &= \tau^{-1}Q_1Dg(0) = \tau^{-1}Q_1t(w) \\ &= \tau^{-1}Q_1t(\text{Proj}_{X_1}(w') + \text{Proj}_{X_2}(w')) = \tau^{-1}Q_1t(\text{Proj}_{X_1}(w')) \\ &= \tau^{-1}Q_1\tau(\text{Proj}_{X_1}(w')) = \tau^{-1}\tau(\text{Proj}_{X_1}(w')) = \text{Proj}_{X_1}(w'); \\ x'_2 &= \text{Proj}_{X_2}(w'), \end{aligned}$$

by linearity of τ^{-1} , Q_1 , by the chain rule and definition of τ , X_1 , X_2 (t vanishes on X_2 and equals τ on X_1). Thus $D_\sigma(0)$ is the identity on X .

We can apply the inverse function theorem to σ near 0. There exists an open neighborhood $W \subset \phi(U) \subset X$ of 0 on which σ is a C^q -diffeomorphism onto $\sigma(W) \subset X$. Let

$$I = I_1 \times I_2$$

be an open interval centered at 0 such that $\bar{I} \subset \sigma(W)$, and such that I_1, I_2 are cubic intervals respectively in X_1 and X_2 . Then

$$(\phi^{-1}\sigma^{-1}(I), \sigma\phi, X)$$

is a chart for M near u_0 , since $\sigma\phi$ is a C^q diffeomorphism of the open set $\phi^{-1}\sigma^{-1}(I)$ onto $I \subset X$ and $u_0 \in \phi^{-1}\sigma^{-1}(I)$.

Consider the C^q map on I into $\psi(V)$ defined as

$$h = g\sigma^{-1} = \psi f \phi^{-1} \sigma^{-1} = \psi f (\sigma\phi)^{-1}.$$

The map h is a local representation of f in terms of the charts $\sigma\phi$ of M and ψ of N . An arbitrary point $u \in \phi^{-1}\sigma^{-1}(I)$ is of rank r relative to f if and only if $\sigma\phi(u) \in I$ is of rank r relative to h .

Let S denote the set of points of I of rank r relative to h . Then $\phi^{-1}\sigma^{-1}(S)$ is precisely the set of points of $\phi^{-1}\sigma^{-1}(I)$ of rank r relative to f ; that is,

$$\phi^{-1}\sigma^{-1}(S) = B_r \cap \phi^{-1}\sigma^{-1}(I).$$

As mentioned in the previous section, to prove that $f(B_r)$ is $(r + \rho)$ -null, it is sufficient to prove that there exists a neighborhood Ω of each $u_0 \in B_r$ such that $f(B_r \cap \Omega)$ is $(r + \rho)$ -null. We intend to prove this for $\Omega = \phi^{-1}\sigma^{-1}(I)$.

By Lemma 1' it is sufficient to show that $h(S)$ is $(r + \rho)$ -null, since

$$h(S) = h\sigma\phi(B_r \cap \Omega) = \psi f(B_r \cap \Omega),$$

and since ψ is a diffeomorphism. The map h carries (x_1, x_2) , $x_1 \in I_1$, $x_2 \in I_2$ into (y_1, y_2) , where

$$\begin{aligned} y_1 &= Q_1 h(x_1, x_2) \in Y_1, \\ y_2 &= Q_2 h(x_1, x_2) \in Y_2. \end{aligned}$$

By (10),

$$Q_1 h(x_1, x_2) = Q_1 g \sigma^{-1}(x_1, x_2) = Q_1 g(w) = \tau(x_1), \quad w \in \sigma^{-1}I \subset W,$$

since σ is a diffeomorphism on W . It follows that h is the map

$$\begin{aligned} y_1 &= \tau(x_1), \\ y_2 &= j(x_1, x_2) := Q_2 h(x_1, x_2) \in Y_2, \end{aligned} \tag{11}$$

where $(x_1, x_2) \in I_1 \times I_2$. The map h is partially linear, and for all $(x_1, x_2) \in I_1 \times I_2$, $\tau(x_1)$ is independent of x_2 . The map j is C^q on the closure of I into Y_2 .

For each $x_1 \in I_1$ and $x_2 \in I_2$, the derivative $Dh(x_1, x_2)$ is the map

$$\begin{aligned} y'_1 &= \tau(x'_1), \\ y'_2 &= D_1 j(x_1, x_2)(x'_1) + D_2 j(x_1, x_2)(x'_2), \end{aligned}$$

of X onto Y , where D_1 and D_2 are the partial derivatives with respect to the first and second argument x_1 and x_2 respectively; $D_1 j(x_1, x_2)$, $D_2 j(x_1, x_2)$ are linear maps of X_1 , X_2 respectively into Y_2 .

Since Y_1 is of dimension r , it follows that for any $(x_1, x_2) \in I$ the image $Dh(x_1, x_2)X$ is r -dimensional if and only if $D_2 j(x_1, x_2) = 0$, hence

$$S = \{(x_1, x_2) \in I : D_2 j(x_1, x_2) = 0\}. \tag{12}$$

This means that $(x_1, x_2) \in I$ is of rank r relative to h if and only if x_2 is of rank 0 relative to the map h with fixed x_1 .

The above analysis applies in a simpler form if $m = r$, since in that case $h = \tau$ and all points of I are of rank m .

This way we have completed the first quarter of the proof, which is the local analysis of f near a point u_0 in the case in which u_0 and $f(u_0)$ are interior points of M, N respectively. If u_0 or $f(u_0)$ were to be boundary points, by the definition of C^q maps we could extend h to a map $\bar{h}$ which is C^q on an open interval of E^m into an entire Banach space. Since the image under $\bar{h}$ contains the image under h , it will be sufficient to work through the same local analysis under $\bar{h}$, so for simplicity we will design $\bar{h}$ as h . Thus the form (11) holds in all cases.

We proceed now to prove that $h(S)$ is $(r + \rho)$ -null.

3.3 Separation of cases

Our goal in this quarter of the proof is to first solve for the case $r = \dim(Y) < \infty$ and then further subdivide S into two sets.

First case. Suppose that $r = \dim(Y) < \infty$. Then Y is an r -dimensional Banach space and is therefore r -sigmafinite. It follows that Y and $h(I)$, $h(S)$ are $(r + \rho)$ -null for all $\rho > 0$ by Lemmas 4 and 2.

Subdivision. We already went through the case $r + \rho > m$ in Section 1, under "Observations", so we have to solve for

$$r < m, \quad r < \dim(Y), \quad 0 < \rho \leq m - r = k. \quad (13)$$

We now divide the set S into the sets Z and T as follows

$$\begin{aligned} Z = & \{(x_1, x_2) \in S : D_2 j(x_1, x_2) = 0 \text{ and } D_2^2 j(x_1, x_2) = 0 \\ & \text{and } \dots \text{ and } D_2^{\nu-1} j(x_1, x_2) = 0\}, \\ T = & S \setminus Z, \end{aligned}$$

with $\nu = \nu(k, \rho)$ as defined in (2). Since $\rho \leq k$ by (13), it follows that $\nu \geq 2$.

If $\nu = 2$, then $Z = S$ by (12) and $T = \emptyset$.

If $\nu > 2$, then Z consists of all points of S at which all the partial derivatives of j with respect to x_2 , of order smaller than ν , vanish. Since

$$h(S) = h(Z) \cup h(T),$$

we now show that $h(Z)$ is $(r + \rho)$ -null (by a covering argument) and $h(T)$ is $(r + \rho)$ -null (by induction).

3.4 $h(Z)$ is $(r + \rho)$ -null

We shall use Taylor's formula [3, §3.3] to obtain some estimates and conclude that $h(Z)$ is $(r + \rho)$ -null by a covering argument.

Estimates. We now prove that there exist constants α_2 and β_1 such that for all $(x_1, x_2) \in Z$, $(x'_1, x'_2) \in I$,

$$\|h(x'_1, x'_2) - h(x_1, x_2)\|_Y \leq \alpha_2 \|x'_1 - x_1\|_{X_1} + \beta_1 \|x'_2 - x_2\|_{X_2}^\nu. \quad (14)$$

To this end, we first prove that there exist constants α_1 and β_1 such that for all $(x_1, x_2) \in Z$, $(x'_1, x'_2) \in I$,

$$\|j(x'_1, x'_2) - j(x_1, x_2)\|_{Y_2} \leq \alpha_1 \|x'_1 - x_1\|_{X_1} + \beta_1 \|x'_2 - x_2\|_{X_2}^\nu. \quad (15)$$

To do so, we use (11), the definition of Z , and the triangle inequality. We take arbitrary $(x_1, x_2) \in Z$, $(x'_1, x'_2) \in I$ and prove the estimate (15) holds. By the triangle inequality,

$$\|j(x'_1, x'_2) - j(x_1, x_2)\|_{Y_2} \leq \|j(x'_1, x'_2) - j(x_1, x'_2)\|_{Y_2} + \|j(x_1, x'_2) - j(x_1, x_2)\|_{Y_2}. \quad (16)$$

Using Taylor's formula on the first term of the right-hand side, it follows that there exists $z \in [x_1, x'_1] \subset I_1$ such that

$$\|j(x'_1, x'_2) - j(x_1, x'_2)\|_{Y_2} = \|D_1 j(z, x'_2)(x'_1 - x_1)\|_{Y_2} \leq \|D_1 j\|_\infty \|x'_1 - x_1\|_{X_1}, \quad (17)$$

where $\|D_1 j\|_\infty < \infty$ since j is a C^q map, with $q \geq 1$. Similarly, for the second term, there exists $w \in [x_2, x'_2] \subset I_2$ such that:

$$\begin{aligned} \|j(x_1, x'_2) - j(x_1, x_2)\|_{Y_2} &= \left\| \sum_{m=1}^{\nu-1} \frac{D_2^m j(x_1, x_2)(x'_2 - x_2, \dots, x'_2 - x_2)}{m!} \right. \\ &\quad \left. + \frac{D_2^\nu j(x_1, w)(x'_2 - x_2, \dots, x'_2 - x_2)}{\nu!} \right\|_{Y_2}. \end{aligned}$$

We note $q \geq c(k, \rho) \geq \nu$ by (4):

$$\left\| \frac{D_2^\nu j(x_1, w)(x'_2 - x_2, \dots, x'_2 - x_2)}{\nu!} \right\|_{Y_2} \leq \frac{\|D_2^\nu j\|_\infty}{\nu!} \|x'_2 - x_2\|_{X_2}^\nu. \quad (18)$$

Combining (16), (17) and (18) we obtain

$$\|j(x'_1, x'_2) - j(x_1, x_2)\|_{Y_2} \leq \|D_1 j\|_\infty \|x'_1 - x_1\|_{X_1} + \frac{\|D_2^\nu j\|_\infty}{\nu!} \|x'_2 - x_2\|_{X_2}^\nu.$$

The estimate (15) follows by defining

$$\alpha_1 = \|D_1 j\|_\infty, \quad \beta_1 = \frac{\|D_2^\nu j\|_\infty}{\nu!}.$$

To prove estimate (14) we use the continuity of τ and the triangle inequality:

$$\begin{aligned}
\|h(x'_1, x'_2) - h(x_1, x_2)\|_Y &= \|j(x'_1, x'_2) - j(x_1, x_2) + \tau(x'_1) - \tau(x_1)\|_Y \\
&\leq \|j(x'_1, x'_2) - j(x_1, x_2)\|_{Y_2} + \|\tau(x'_1) - \tau(x_1)\|_{Y_1} \\
&\leq \alpha_1 \|x'_1 - x_1\|_{X_1} + \beta_1 \|x'_2 - x_2\|_{X_2}^\nu + \|\tau\|_\infty \|x'_1 - x_1\|_{X_1}.
\end{aligned}$$

We conclude by defining $\alpha_2 = \alpha_1 + \|\tau\|_\infty = \|D_1 j\|_\infty + \|\tau\|_\infty$.

Covering argument. For the covering argument part, we shall divide $I = I_1 \times I_2$ into small subintervals of the same shape. We will then use the estimate above (14) to bound the diameters of the intervals, and then the $r + \rho$ -dimensional Hausdorff measure of $h(Z)$.

Let a be the length of the edge of the cube I_1 , b that of I_2 . We divide each edge of I_1 into 2^λ equal parts, and the edge of I_2 into 2^γ equal parts, where λ and γ are positive integers to be specified later. Then, I is divided into

$$2^{\lambda r + \gamma k}$$

congruent subintervals, as $\dim(I_1) = r$ and $\dim(I_2) = m - r = k$. All of the intervals have each edge either parallel to an edge of I_1 with length $\frac{a}{2^\lambda}$, or of I_2 with length $\frac{b}{2^\gamma}$.

By (14) the image under h of every interval, intersected Z , has diameter δ such that

$$\begin{aligned}
\delta &= \sup_{(x'_1, x'_2), (x_1, x_2) \in J \cap Z} \|h(x'_1, x'_2) - h(x_1, x_2)\|_Y \\
&\leq \sup_{(x'_1, x'_2), (x_1, x_2) \in J \cap Z} \alpha_2 \|x'_1 - x_1\|_{X_1} + \beta_1 \|x'_2 - x_2\|_{X_2}^\nu \\
&\leq \alpha_2 \frac{a}{2^\lambda} + \beta_1 \frac{b^\nu}{2^{\gamma\nu}}.
\end{aligned}$$

For simplicity, we will write $\alpha = \alpha_2 a$ and $\beta = \beta_1 b^\nu$, obtaining

$$\delta \leq \frac{\alpha}{2^\lambda} + \frac{\beta}{2^{\gamma\nu}}.$$

The images of the subintervals defined above, each intersected with Z , constitute a cover of hZ , for which

$$\sum \delta^{r+\rho} \leq 2^{\lambda r + \gamma k} \left(\frac{\alpha}{2^\lambda} + \frac{\beta}{2^{\gamma \nu}} \right) = \left(\frac{\alpha}{2^e} + \frac{\beta}{2^{e'}} \right)^{r+\rho},$$

where

$$\begin{aligned} e &= \lambda - \frac{\lambda r + \gamma k}{r + \rho} = \lambda \frac{\rho}{r + \rho} - \gamma \frac{k}{r + \rho} \\ e' &= \gamma \nu - \frac{\lambda r + \gamma k}{r + \rho} = -\lambda \frac{r}{r + \rho} - \gamma \left(\nu - \frac{k}{r + \rho} \right). \end{aligned} \quad (19)$$

Now, by (2)

$$\nu > \frac{k}{\rho}.$$

This implies that the coefficient of γ in (19) is positive; ergo, we may choose a sequence $\{(\lambda_n, \gamma_n)\}_{n \in \mathbb{N}} \subset \mathbb{R}_{>0} \times \mathbb{R}_{>0}$ such that

$$(\lambda_n, \gamma_n) \xrightarrow{n \rightarrow \infty} (\infty, \infty), \quad e \rightarrow \infty, \quad e' \rightarrow \infty,$$

since there exists a sector of the positive angle in the first quadrant of the λ, γ -plane. Since $\alpha, \beta, r + \rho$ are constants, it follows that $h(Z)$ is $(r + \rho)$ -null.

3.5 $h(T)$ is $(r + \rho)$ -null

We now move to the last part of the proof. We will first apply Lemma 6 and we will then set up an induction.

Application of Lemma 6. We have seen in the paragraph “Subdivision” that $T = \emptyset$ if $\nu \geq 2$. It is sufficient then to consider $h(T)$ in the case

$$\nu \geq 3, \quad 0 < \rho \leq k,$$

by (13). In particular $k \geq 1$, being k an integer.

The set T consists of points $(x_1, x_2) \in I$ at which $D_2 h(x_1, x_2) = 0$, but not all maps $D_2^2 h(x_1, x_2), D_2^3 h(x_1, x_2), \dots, D_2^{\nu-1} h(x_1, x_2)$ vanish. We divide T into $\nu - 2$ sets as follows:

$$T_i = \{(x_1, x_2) \in S : D_2 h(x_1, x_2) = 0 \text{ and } \dots \\ \text{and } D_2^i h(x_1, x_2) = 0 \text{ and } D_2^{i+1} h(x_1, x_2) \neq 0\}, \quad i = 1, 2, \dots, \nu - 2.$$

Then

$$h(T) = h(T_1) \cup h(T_2) \cup \dots \cup h(T_{\nu-2}).$$

It will be sufficient to show that each term is $(r + \rho)$ -null.

Consider $h(T_i)$ with i fixed and $1 \leq i \leq \nu - 2$. To prove that $h(T_i)$ is $(r + \rho)$ -null it will be sufficient to reuse a strategy we have already used: we show that each point $x_0 \in T_i$ lies in an open set Ω such that $h(T_i \cap \Omega)$ is $(r + \rho)$ -null, since countably many such sets will cover T_i .

By Lemma 6 with $\theta = D_2^i h$, there is an open neighborhood Ω of $x_0 \in T_i$ in which one coordinate of x , say x^m , satisfies a relation

$$x^m = \eta(x^1, \dots, x^{m-1}), \text{ whenever } x = (x^1, \dots, x^m) \in T_i \cap \Omega,$$

where η is a $C^{q-\nu+2}$ map of an open neighborhood $\tilde{\Omega}$ of $(x_0^1, \dots, x_0^{m-1}) \in E^{m-1}$. The stated class of η is assured because $D_2^i h$ is a C^{q-i} map of I into a Banach space of maps and because $i \leq \nu - 2$. It follows that $h(T_i \cap \Omega)$ is contained in $\tilde{h}(\tilde{\Omega})$, where $\tilde{\Omega}$ is as defined in Lemma 6, and

$$\tilde{h}(x_1, \dots, x_{m-1}) = h(x_1, \dots, x_{m-1}, \eta(x_1, \dots, x_{m-1})), \quad (x_1, \dots, x_{m-1}) \in \tilde{\Omega}$$

Moreover, $h(T_i \cap \Omega)$ is contained in the image under $\tilde{h}$ of the points of $\tilde{\Omega}$ of rank $\leq r$, because the rank of $(x_1, \dots, x_{m-1})$ relative to h is at most the rank of $(x_1, \dots, x_{m-1}, \eta(x_1, \dots, x_{m-1}))$ relative to h .

It follows that $h(T_i \cap \Omega)$ is surely $(r + \rho)$ -null if $h(\tilde{S})$ is $(r + \rho)$ -null, where $\tilde{S}$ consists of the points of $\tilde{h}$ of rank $\leq r$.

Induction. We now reason by induction on m ¹: for $m = 0$ the theorem follows immediately, since M is countable.

¹Similar induction have been used by A.B. Brown [2], M. Morse Sard [18] and Pontryagin [21]

Consider the case $m = 1$. The reduction of dimension implies that x_0 is isolated in $T_i \cap \Omega$, since the function η becomes constant. Thus $h(T_i \cap \Omega)$ consists of a point and is therefore $(r + \rho)$ -null. This proves the theorem for $m = 1$.

We now assume the theorem has been proved for maps on manifolds of dimension $m - 1$ into N . Then $\tilde{h}\tilde{S}$ is surely $(r + \rho)$ -null, since $\tilde{h}$ is such a map and since

$$q - \nu + 2 \geq c(k - 1, \rho),$$

by (5). Hence $h(T_i \cap \Omega)$ is $(r + \rho)$ -null, and the theorem is established for maps on manifolds of dimension m .

In particular, if f is a C^∞ map, then $f(A_r)$ is $(r + \rho)$ -null for all $\rho > 0$, which implies that $\dim_{\mathcal{H}}(A_r) \leq r$. $\square$

Chapter 4

Further topics

In this chapter, we shall discuss several statements; seven are proved theorems, and one - a Fubini Theorem generalization - we will prove to be false.

Throughout the chapter, we will keep the notations established in Chapter 3 and refer to different parts of the proof to think about what could be different under changes in the hypotheses.

Excluding Theorem 2, we will work under the assumption that N is finite-dimensional.

First considerations. The theorem is satisfactory in several ways: ρ may be any positive number; the manifold N is very general and the dimensionality in particular does not enter the condition on q . The hypothesis

$$q \geq c(k, \rho)$$

however, is duly demanding. It is in our interest to discuss the difficulties in the way of obtaining stronger theorems.

4.1 Some further results

We will now present seven further results, all related to the theorem we proved in Chapter 3.

Theorem 2. A point $u \in M$ is called simple if f has constant rank in some neighborhood of u . Let K denote the set of simple points of M . Then the image

$$f(A_r \cap K)$$

of simple points of rank at most r is r -sigmafinite. No condition other than $q \geq 1$ is needed.

Proof. It suffices to show that $f(B_r \cap K)$ is r -sigmafinite. If $u_0 \in B_r \cap K$, then the neighborhood $I_1 \times I_2$ defined in Section 3.2 may be taken so that $D_2 j(x_1, x_2)$ depends on x_1 alone, so that

$$j(x_1, x_2) = j(x_1), \quad x_1 \in I_1, \quad x_2 \in I_2.$$

Introduce the new chart $\omega \circ \psi$ for N , where ω is the diffeomorphism

$$\begin{aligned}(z_1, z_2) &= \omega(y_1, y_2) \\ z_1 &= y_1, & y_1 &\in \tau(I_1), \\ z_2 &= y_2 - j\tau^{-1}(y_1), & y_2 &\in Y_2.\end{aligned}$$

Then the local representation of the map f in terms of the charts $\sigma \circ \phi$ and $\omega \circ \psi$ is

$$\begin{aligned}z_1 &= \tau(x_1), & x_1 &\in I_1 \\ z_2 &= 0\end{aligned}$$

This map is entirely linear. Its image is a parallelotope in E^r and is therefore r finite. Since $B_r \cap K$ may be covered by countably many neighborhoods $\phi^{-1} \circ \sigma^{-1}(I_1 \times I_2)$, it follows that $f(B_r \cap K)$ is r -sigmafinite, as was to be shown. $\square$

From now on, assume that N is finite-dimensional. Thus f is a map of an m -dimensional manifold M into an n -dimensional manifold N , $1 \leq m, n < \infty$; M, N, f are C^q , $1 \leq q \leq \infty$; M has countable basis; A_r denotes the set of points of M of rank at most r relative to f .

Theorem 3. $f(A_0)$ is ρ -null if $q \geq \frac{m}{\rho}$, $\rho > 0$.

See [22, Th. 6.1]. The proof is based on Morse's theorem on the decomposition of critical sets [17].

Theorem 4. $f(A_{m-1})$ is m -null if $q \geq 1$.

Knopp and Schmidt established Theorem 4 for the case $m = n$ in 1962 [14], and the general version appeared in 1958 [23].

Theorem 5. $f(A_{n-1})$ is n -null if $q \geq m - n + 1$; however $f(A_{n-1})$ may be n -positive if $q = m - n$.

See [22, Th. 7.2]. The proof depends on Theorem 3 and the theorem of Fubini.

The papers [5], [6] and [7] of Dubovickii deal with results related to Theorem 3 and Theorem 5.

Theorem 6. $f(A_r)$ is $(r + \rho)$ -null if $q > \frac{m - r}{\rho}$, $\rho > 0$.

See [11] for a proof. The inequality is sharp.

Theorem 7. $f(A_r)$ is $(r + \rho)$ -null if $q > m - r$.

Again, see [11] for a proof.

Definition (refinements). Let (X, τ_X) be a topological space. A cover $\mathcal{V}$ is a refinement of another cover $\mathcal{U}$ if for every $V \in \mathcal{V}$ there exist $U \in \mathcal{U}$ with $V \subset U$.

Definition (topological order of a cover). Let (X, τ_X) be a topological space, $\mathcal{U}$ an open cover. The order of $\mathcal{U}$ is defined as

$$m(\mathcal{U}) = \sup_{x \in X} \# \{ U \in \mathcal{U} \mid x \in U \}.$$

Definition (topological dimension). The topological dimension or Lebesgue covering dimension of X , denoted $\dim_X$, is the least integer $n \geq 0$ such that for every open cover $\mathcal{U}$, there exists a refinement $\mathcal{V}$ with order at most $n + 1$. If no such set exists, then $\dim X = \infty$. By convention, $\dim \emptyset = -1$.

Remark. As we did for Hausdorff dimension, let us state some properties of the topological dimension. Let (X, τ_X) , (Y, τ_Y) be topological spaces:

- if $A \subset B \subset X$, then $\dim(A) \leq \dim(B)$;
- if $A \subset \mathbb{R}^m$, then $\dim(U) = m$ for every open $\emptyset \neq U \subset \mathbb{R}^m$;
- if X is homeomorphic to Y , then $\dim(X) = \dim(Y)$;
- if N is an n -dimensional topological manifold, then $\dim(N) = n$;
- if $X = A \cup B$ with A, B closed, then $\dim(X) \leq \max\{\dim(A), \dim(B)\}$;
- if $X = \bigcup_{i=1}^{\infty} F_i$ with each F_i closed, then $\dim(X) = \sup_i \dim(F_i)$.

These facts are standard; we will neither prove nor use them. To know more, see [13].

Theorem 8. $f(A_r)$ is of topological dimension $\leq r$ if $q \geq m - r$.

The theorem follows from a theorem of Szpilrajn (Marezewski) (see [13]).

4.2 One counterexample

Now consider the question, partially answered by Theorem 1: when may we assert that $f(A_r)$ is $(r + \rho)$ -null for specified $\rho > 0$? Proceeding as in Chapter 3, we see that it would be sufficient to show that $h(S)$ is $(r + \rho)$ -null, where h is the map (11) and S is the set (12).

We saw in Section 3.2 that $(x_1, x_2) \in S$ if and only if x_2 is of rank 0 relative to the map h with x_1 fixed of an open set in E^k into E^{n-k} . By (11), the cross section of $h(S)$ obtained by fixing $y_1 = \tau(x_1)$, $x_1 \in X_1$, is itself the image under h , with x_1 fixed, of its points of rank 0.

By Theorem 3, the cross section of $h(S)$, y_1 fixed, is ρ -null if $q \geq \frac{k}{\rho}$, for all $y_1 \in Y_1$, since the map h with x_1 fixed has k -dimensional domain.

Now, an important fact in the theory of Hausdorff measure is that the following generalization of Fubini's theorem is not valid.

Statement 1. Consider a compact set $B \subset E^r \times E^{n-r}$, $1 \leq r \leq n - 1$. Let B_{y_1} denote the cross section

$$\{y_2 : (y_1, y_2) \in B\} \subset E^{n-r}$$

of B at $y_1 \in E^r$. Suppose that B_{y_1} is ρ -null for each $y_1 \in E^r$. Then B is $(r + \rho)$ -null.

This statement has been shown to be false by Federer in 1964, who built the following counterexample.

Definition (Peano map). A map from the unit square in E^n onto the unit square in E^m , $n, m \in \mathbb{N}$, $n, m \geq 1$ is a Peano map if it is surjective and continuous.

Example (Hilbert curve). The Hilbert curve (see [12]) is an example of Peano map from $[0, 1]$ onto $[0, 1] \times [0, 1]$.

It is defined as follows: take $[0, 1]$, and first divide it into four equal parts, segments 1, 2, 3, and 4; divide the surface of the square by two perpendicular lines into four equal subsquares 1, 2, 3, 4 (see Figure 1.a)).

Next, subdivide each of the four segments 1, 2, 3, and 4 again into four equal parts each, so that on the line we obtain 16 subsegments 1, 2, 3, ..., 16; at the same time each of the four squares is divided into four equal subsquares, and the numbers 1, 2, ..., 16 are assigned to them, choosing the order so that each successive square

touches the preceding one along a side (see Figure 1.b).

Continuing this procedure (see (see Figure 1.c)), it is easy to assign to each given point of the segment a single, definite point of the square: one need only determine in which subsegments of the line the given point lies; the squares carrying the same numbers are nested and in the limit enclose a definite point of the square.

The map obtained in this way is single-valued and continuous; conversely, to each point of the square there correspond at most four points of the line.

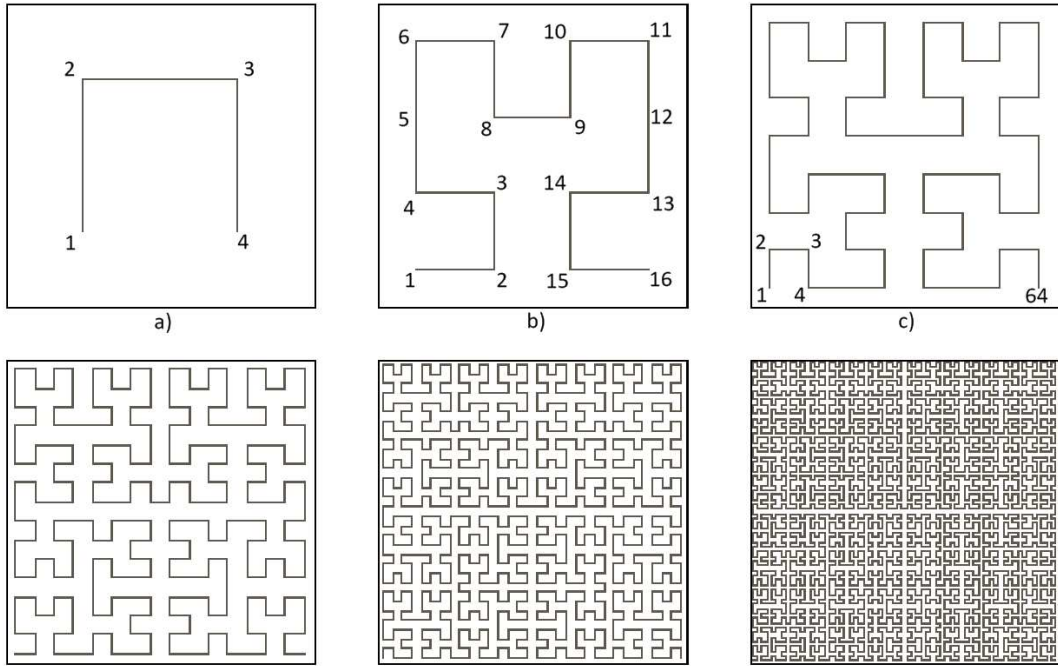


Figure 1: Hilbert's curve, first six iterations

Observation 2. There exists at least one Peano map from the unit square in E^n onto the unit square in E^m (either a projection if $m \leq n$ or an application of Hilbert's curve on $m - n$ components of E^n).

Counterexample 1. Let p be a Peano map, single valued and continuous, of the unit square in E^r onto the entire square in E^{n-r} . Put

$$B = \{(y_1, y_2) \in E^n \mid y_2 = p(y_1), y_1 \in E^r\}.$$

Then B_{y_1} is, by construction, either a single point or the empty set. Therefore B_{y_1} is ρ -null, for any $\rho > 0$, for all $y_1 \in E^r$. Nonetheless B is $(r + \rho)$ -positive if $r + \rho \leq n - r$; this is because

$$\mathcal{H}^{n-r}(B) \geq \mathcal{H}^{n-r} \text{Proj}_{E^{n-r}} B > 0,$$

since the projection of B on E^{n-r} is the unit square. Thus statement 1 is false.

Remark. Statement 1 is true if $\rho = n - r$, by Fubini's theorem; it is also true if B_{y_1} is independent of y_1 (see [10, Lemma 4.1]).

Bibliography

- [1] Abram S. Besicovitch. A problem on measure. *Proc. Cambridge Philos. Soc.*, 59:251–253, 1963.
- [2] A. B. Brown. Functional dependence. *Trans. Amer. Math. Soc.*, 38:379–394, 1935.
- [3] Rodney Coleman. *Calculus on Normed Vector Spaces*. Universitext. Springer, New York, NY, 2012. eBook ISBN: 978-1-4614-3894-6.
- [4] Jean Dieudonné. *Foundations of Modern Analysis*. Number 10 in Pure and Applied Mathematics. Academic Press, New York, 1960.
- [5] A. Ya. Dubovickii. On differentiable mappings of an n -dimensional cube into a k -dimensional cube. *Mat. Sb. (N.S.)*, 32 (74):443–464, 1953. In Russian.
- [6] A. Ya. Dubovickii. On the structure of level sets of differentiable mappings of an n -dimensional cube into a k -dimensional cube. *Izv. Akad. Nauk SSSR Ser. Mat.*, 21:371–408, 1957. In Russian.
- [7] A. Ya. Dubovickii. Points of complete degeneracy of the jacobi matrix. *Izv. Akad. Nauk SSSR Ser. Mat.*, 22:705–716, 1958. In Russian.
- [8] Nelson Dunford and Jacob T. Schwartz. *Linear Operators, Part I: General Theory*. Number 7 in Interscience Monographs in Pure and Applied Mathematics. Interscience Publishers, New York, 1958.
- [9] K. J. Falconer. *The Geometry of Fractal Sets*. Number 85 in Cambridge Tracts in Mathematics. Cambridge University Press, Cambridge, 1985.
- [10] Herbert Federer. Some integral geometric theorems. *Trans. Amer. Math. Soc.*, 77:238–261, 1954.
- [11] Herbert Federer. *Geometric Measure Theory*. Number 153 in Grundlehren der Mathematischen Wissenschaften. Springer, Berlin, 1969. See Chapter 3 for the Dubovickii–Federer estimate.
- [12] David Hilbert. Über die stetige abbildung einer linie auf ein flächenstück. *Math. Ann.*, 38:459–460, 1891. In German.
- [13] Witold Hurewicz and Henry Wallman. *Dimension Theory*. Number 4 in Princeton Mathematical Series. Princeton University Press, Princeton, NJ, 1941.

- [14] Konrad Knopp and Rudolf Schmidt. Funktionaldeterminanten und abhängigkeit von funktionen. *Math. Z.*, 25:373–381, 1926.
- [15] Ivan Kupka. Counterexample to the Morse–Sard theorem in the case of infinite-dimensional manifolds. *Proc. Amer. Math. Soc.*, 16(5):954–957, 1965.
- [16] Serge Lang. *Introduction to Differentiable Manifolds*. Interscience Publishers, New York, 1962.
- [17] Anthony P. Morse. The behavior of a function on its critical set. *Ann. of Math.*, 40(1):62–70, 1939.
- [18] Marston Morse and Arthur Sard. On the measure of the critical values of functions. Unpublished manuscript, 1935.
- [19] James R. Munkres. *Elementary Differential Topology*. Number 54 in Annals of Mathematics Studies. Princeton University Press, Princeton, NJ, 1963.
- [20] Richard S. Palais. Morse theory on Hilbert manifolds. *Topology*, 2(4):299–340, 1963.
- [21] L. S. Pontryagin. Smooth manifolds and their applications in homotopy theory. *Amer. Math. Soc. Transl. (2)*, 11:1–114, 1959. Translated from the Russian: *Trudy Mat. Inst. Steklov.*, no. 45 (1955).
- [22] Arthur Sard. The measure of the critical values of differentiable maps. *Bull. Amer. Math. Soc.*, 48(12):883–890, 1942.
- [23] Arthur Sard. Images of critical sets. *Ann. of Math.*, 68:247–259, 1958.
- [24] Arthur Sard. Hausdorff measure of critical images on Banach manifolds. *Amer. J. Math.*, 87(1):158–174, 1965.
- [25] Hassler Whitney. A function not constant on a connected set of critical points. *Duke Math. J.*, 1(4):514–517, 1935.